

DMS-613 Project Report

Solvency Cones and Superhedging in Markets with Transaction Costs

Aryan Chaturvedi (230212)

Tanay Pathode (231074)

1 Introduction

In financial markets, transaction costs such as bid–ask prices play a crucial role in trading and pricing. Ideal frictionless models assume perfect liquidity and no trading costs, leading to linear pricing and replication-based hedging. However, these assumptions are not used in practice.

In this paper we study a mathematical framework for modeling markets with proportional transaction costs. Within this framework, a set-valued approach to superhedging is used, where instead of determining a single price, the entire set of initial portfolios that can hedge a given contingent claim is calculated.

A central concept in this approach is the solvency cone, which describes all feasible trades in the market and determines whether a portfolio can be transformed into a solvent (non-negative) position. Based on this structure, a recursive scheme is derived for computing the set of superhedging portfolios.

2 Problem Statement

We consider a discrete-time financial market with multiple assets and proportional transaction costs.

The main objective is:

To determine all initial portfolios that can superhedge a given contingent claim.

Unlike classical models, this problem does not lead to a single price. Instead, the solution is a set of portfolios that guarantee hedging under all possible future scenarios.

This shift from scalar pricing to set-valued hedging is the central theme of this work.

3 Market Model

We consider a discrete-time market with d assets over time $t = 0, 1, \dots, T$, defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

3.1 Portfolios

A portfolio at time t is represented by a vector

$$V_t \in \mathbb{R}^d,$$

where each component denotes the number of units held in a particular asset.

For each state ω , the portfolio can be written as

$$V_t(\omega) = (v_1, v_2, \dots, v_d),$$

which shows that the portfolio may vary across different states of the world.

We note that V_t is $\mathcal{F}_t$ -measurable, meaning that the portfolio at time t depends only on the information available up to that time.

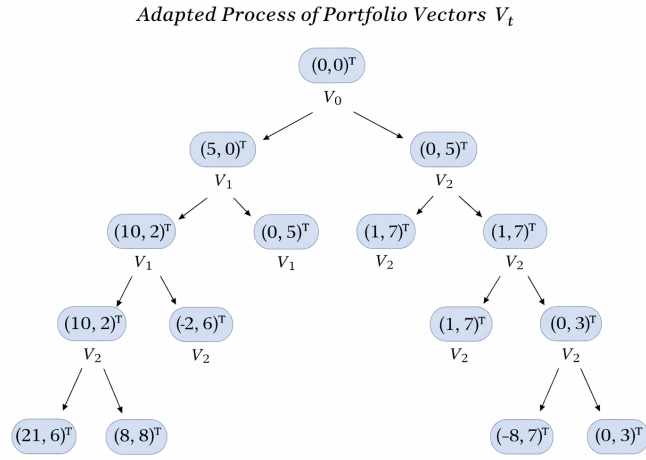


Figure 1: Demo Portfolio evolution across states

3.2 Bid-Ask Matrix

Trading is governed by a bid-ask matrix $\Pi_t = (\pi_t^{ij})$, where:

$$\pi_t^{ij} = \text{units of asset } i \text{ needed to obtain 1 unit of asset } j.$$

This matrix encodes both exchange rates and transaction costs in the market.

The pair

$$\left(\frac{1}{\pi^{ji}}, \pi^{ij} \right)$$

represents the bid and ask prices of asset j in terms of asset i .

The matrix satisfies:

- $\pi^{ij} > 0$ (prices are positive),
- $\pi^{ii} = 1$ (no cost for self-exchange),
- $\pi^{ij} \leq \pi^{ik} \pi^{kj}$ (no arbitrage through indirect trades).

4 Solvency Cones

4.1 Definition and Structure

At each time $t \in \{0, 1, \dots, T\}$ and state $\omega \in \Omega$, the *solvency cone* $K_t(\omega) \subseteq \mathbb{R}^d$ is defined as a polyhedral convex cone generated by the vectors

$$\pi^{ij}(\omega)e_i - e_j, \quad 1 \leq i, j \leq d,$$

together with the unit vectors

$$e_i, \quad 1 \leq i \leq d.$$

Thus, the cone admits the representation

$$K_t(\omega) = \text{cone} \left\{ \pi^{ij}(\omega)e_i - e_j, e_i \mid i, j = 1, \dots, d \right\}.$$

Here, each vector $\pi^{ij}(\omega)e_i - e_j$ corresponds to an exchange of one unit of asset j for $\pi^{ij}(\omega)$ units of asset i , while the vectors e_i represent the possibility of free disposal of assets. Hence as it is spanned by the negative of trade vectors, it contains negative of all available trades in markets. Hence all trades at time t are contained in $-K_t$.

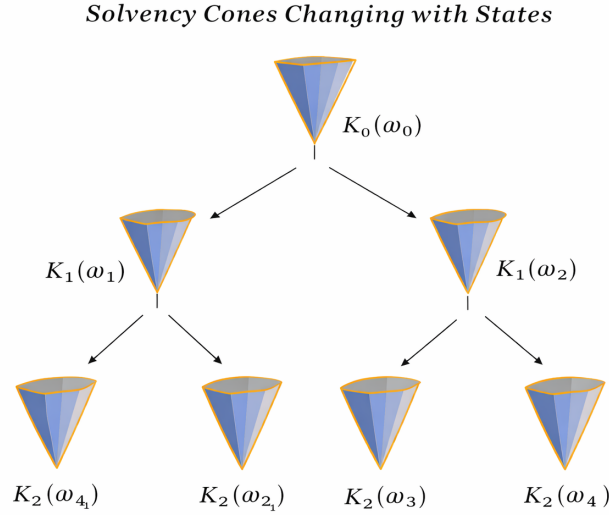


Figure 2: Dummy example - Cones evolution across states

4.2 Liquidity Assumption

Throughout the model, we assume that

$$\mathbb{R}_+^d \setminus \{0\} \subseteq \text{int } K_t(\omega).$$

This condition ensures that the interior of the solvency cone contains all strictly positive portfolios. Financially, this corresponds to a liquidity assumption, meaning that exchange rates between any pair of assets are strictly positive and finite.

An important implication of this assumption is that any asset can serve as a numéraire, which is useful when comparing this framework with scalar pricing approaches. While this assumption simplifies the analysis, it is not required for all theoretical results.

4.3 Financial Interpretation

The cone $K_t(\omega)$ is called the *solvency cone* because it characterizes precisely those portfolios that can be transformed into non-negative portfolios through trading at time t .

More precisely, a portfolio $x \in \mathbb{R}^d$ belongs to $K_t(\omega)$ if and only if there exists $v \in -K_t(\omega)$ such that

$$x + v \in \mathbb{R}_+^d.$$

Thus, $K_t(\omega)$ represents the set of portfolios that can be made solvent.

4.4 Geometric and Trading Interpretation

The generators of $K_t(\omega)$ represent basic trading directions:

- $\pi^{ij}e_i - e_j$ corresponds to the exchange of asset j for asset i ,
- e_i represents the disposal of asset i .

Since $K_t(\omega)$ is a cone, it consists of all non-negative combinations of these generators. This reflects the fact that trades can be scaled and combined, but not freely reversed due to transaction costs. As a result, the set of feasible trades forms a cone rather than a linear space.

4.5 Role in the Model

The solvency cone helps us understand what trades are actually possible in the market when transaction costs are present. It ensures that any change in the portfolio comes only from valid trades, as given by the self-financing condition

$$V_t - V_{t-1} \in -K_t.$$

From our understanding, it captures both exchanging assets and discarding them when needed. Overall, it acts as the basic structure on which attainable portfolios and superhedging are built.

5 Spaces of Random Portfolios and Self-Financing Trading

5.1 Spaces of Measurable Random Vectors

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0}^T, \mathbb{P})$ be a filtered probability space.

For each $t \in \{0, 1, \dots, T\}$, we define

$$L^0(\mathcal{F}_t, \mathbb{R}^d)$$

as the space of all $\mathcal{F}_t$ -measurable random vectors

$$X : \Omega \rightarrow \mathbb{R}^d.$$

This space represents all portfolios whose values are known at time t .

More generally, for a set $D_t \subseteq \mathbb{R}^d$, we define

$$L^0(\mathcal{F}_t, D_t)$$

as the set of all $\mathcal{F}_t$ -measurable random vectors that take values in D_t almost surely.

Since we work with a finite probability space, each random vector can be identified with a finite-dimensional vector. In particular, we may identify

$$L^0(\mathcal{F}_T, \mathbb{R}^d) \cong \mathbb{R}^{dN},$$

where N is the number of possible states in Ω .

These spaces provide the appropriate framework for describing portfolios and trading strategies in a stochastic setting, ensuring that all quantities are adapted to the available information.

5.2 Portfolio Processes

A portfolio process is a sequence of random vectors

$$(V_t)_{t=0}^T,$$

where for each $t \in \{0, 1, \dots, T\}$,

$$V_t \in L^0(\mathcal{F}_t, \mathbb{R}^d).$$

Thus, $V_t(\omega)$ represents the holdings in each asset at time t in state ω , and the $\mathcal{F}_t$ -measurability ensures that the portfolio depends only on the information available at time t .

5.3 Self-Financing Condition

A portfolio process $(V_t)_{t=0}^T$ is called *self-financing* if

$$V_t - V_{t-1} \in -K_t \quad \text{a.s. for all } t \in \{0, \dots, T\},$$

with the convention $V_{-1} = 0$.

Interpretation: This condition ensures that changes in the portfolio are achieved only through trading in the market. In particular, it reflects that all adjustments must lie within the set of feasible trades, and no external capital is injected.

This can be understood as:

All changes in the portfolio must come from market trades.

5.4 Attainable Portfolios

We now define the set of attainable terminal portfolios.

Let

$$A_T \subseteq L^0(\mathcal{F}_T, \mathbb{R}^d)$$

denote the set of all random vectors V_T that can be obtained as the terminal value of a self-financing portfolio process starting from zero initial endowment.

Then A_T admits the representation

$$A_T = -L^0(\mathcal{F}_0, K_0) - L^0(\mathcal{F}_1, K_1) - \dots - L^0(\mathcal{F}_T, K_T).$$

Interpretation: Each term $-L^0(\mathcal{F}_t, K_t)$ represents all feasible trades that can be carried out at time t , depending on the available information. The sum over all time steps describes all portfolios that can be reached through sequential trading starting from zero initial capital.

It follows directly from the definition that A_T is a convex cone. This reflects the fact that attainable portfolios can be combined and scaled, consistent with the cone structure of the solvency cones.

Financial Meaning: The set A_T represents all claims that can be superhedged starting from zero initial endowment. It plays a central role in the formulation of the superhedging problem.

6 No Arbitrage and Pricing

The market is said to satisfy the weak no arbitrage condition if:

$$A_T \cap L^0(\mathcal{F}_T, \mathbb{R}_+^d) = \{0\}.$$

This means that it is not possible to generate a non-negative portfolio with strictly positive gain starting from zero initial capital.

The dual object associated with solvency cones is the dual cone:

$$K_t^+(\omega) = \{v \in \mathbb{R}^d \mid v^T u \geq 0 \ \forall u \in K_t(\omega)\}.$$

This represents all price systems that are consistent with transaction costs.

Fundamental Theorem of Asset Pricing:

The market satisfies no arbitrage if and only if there exists a consistent price system (Z_t) such that:

- $Z_t \in K_t^+$,
- Z_t is a martingale.

Thus, instead of a single price, we obtain a set of price systems that respect transaction costs.

7 Superhedging

7.1 Definition

An initial portfolio x_0 is said to superhedge a claim X if:

$$X \in x_0 + A_T.$$

Understanding: This means that starting from x_0 , we can construct a self-financing trading strategy whose terminal value dominates the claim X in every possible state of the world.

Unlike classical models, superhedging does not require exact replication. Instead, it allows for excess, meaning that the final portfolio may be strictly larger than the claim.

7.2 Superhedging Set

The set of all superhedging portfolios is given by:

$$SHP_0(X) = \{x_0 \in \mathbb{R}^d : X \in x_0 + A_T\}.$$

Understanding: In markets with transaction costs, there is no unique hedging price. Instead, we obtain a set of portfolios, each of which is sufficient to hedge the claim. This reflects the fact that different combinations of assets can achieve the same hedging objective.

8 Dual Representation

The set of superhedging portfolios admits a dual characterization in terms of consistent price systems and vector probability measures.

8.1 Consistent Price Systems

An $\mathbb{R}_+^d$ -valued adapted process $Z = (Z_t)_{t=0}^T$ is called a *consistent price system* for the market model $(K_t)_{t=0}^T$ if:

- Z is a martingale under $\mathbb{P}$,
- $Z_t \in K_t^+ \setminus \{0\}$ almost surely for all $t \in \{0, 1, \dots, T\}$,

where K_t^+ denotes the positive dual cone of K_t .

Let $\mathcal{Z}$ denote the set of all consistent price systems.

Interpretation: A consistent price system represents a generalized pricing rule under transaction costs. The condition $Z_t \in K_t^+$ ensures compatibility with the bid–ask structure, while the martingale property reflects the absence of arbitrage.

8.2 Dual Variables

Define the set of dual elements

$$\mathcal{W}_1 = \left\{ (Q, w) \in \mathcal{M}_{1,d}^P \times \mathbb{R}^d \setminus \{0\} \mid \mathbb{E} \left[\text{diag}(w) \frac{dQ}{d\mathbb{P}} \mid \mathcal{F}_t \right] \in L^1(\mathcal{F}_t, K_t^+) \right\}.$$

Here, $\mathcal{M}_{1,d}^P$ denotes the set of vector probability measures whose components are absolutely continuous with respect to $\mathbb{P}$.

Structure of the Dual Elements:

Each element $(Q, w) \in \mathcal{W}_1$ consists of:

- A *vector probability measure*

$$Q = (Q_1, Q_2, \dots, Q_d),$$

where each Q_i is a probability measure on $(\Omega, \mathcal{F}_T)$ such that

$$\frac{dQ_i}{d\mathbb{P}} \in L^1(\mathcal{F}_T).$$

This allows each asset to be evaluated under its own probability weighting.

- A non-zero vector

$$w = (w_1, w_2, \dots, w_d) \in \mathbb{R}^d \setminus \{0\},$$

which defines a linear functional on portfolios via $w^T x$. This vector specifies a direction in which portfolios are compared.

- The term

$$\text{diag}(w) \frac{dQ}{d\mathbb{P}} = \left(w_1 \frac{dQ_1}{d\mathbb{P}}, \dots, w_d \frac{dQ_d}{d\mathbb{P}} \right),$$

which combines the weighting vector w with the Radon–Nikodym derivatives of the measures.

The condition

$$\mathbb{E} \left[\text{diag}(w) \frac{dQ}{d\mathbb{P}} \middle| \mathcal{F}_t \right] \in K_t^+$$

ensures that the resulting process is consistent with the dual cone of the solvency cone, and therefore respects the bid–ask structure of the market.

Interpretation:

The pair (Q, w) provides a dual description of pricing systems in markets with transaction costs. The measure Q captures probabilistic valuation of future payoffs, while the vector w determines how different assets are aggregated into a scalar value.

Together, they define a pricing rule of the form

$$x \mapsto w^T \mathbb{E}^Q[x],$$

which assigns a scalar value to a vector-valued portfolio.

The condition defining $\mathcal{W}_1$ ensures that these pricing rules are compatible with the market frictions encoded by the solvency cones.

Moreover, the set $\mathcal{W}_1$ is in one-to-one correspondence with the set of consistent price systems $\mathcal{Z}$, providing an equivalent dual representation of the market.

8.3 Dual Representation of Superhedging

Under the no-arbitrage condition, the set of superhedging portfolios of a claim $X \in L^0(\mathcal{F}_T, \mathbb{R}^d)$ satisfies:

$$SHP_0(X) = \{x_0 \in \mathbb{R}^d \mid \forall Z \in \mathcal{Z} : \mathbb{E}[X^T Z_T] \leq x_0^T Z_0\}.$$

Interpretation: This representation states that an initial portfolio x_0 is sufficient to superhedge X if its value dominates the expected value of the claim under all consistent price systems. Thus, superhedging is characterized by a family of pricing inequalities rather than a single price.

8.4 Equivalent Representation via Vector Measures

An equivalent representation is given by:

$$SHP_0(X) = \bigcap_{(Q,w) \in \mathcal{W}_1} (\mathbb{E}^Q[X] + G(w)),$$

where

$$G(w) = \{x \in \mathbb{R}^d \mid w^T x \geq 0\}.$$

Interpretation: Each pair (Q, w) defines a half-space constraint on admissible initial portfolios. The set of superhedging portfolios is obtained as the intersection of all such half-spaces. This provides a geometric characterization of $SHP_0(X)$ as a convex set.

8.5 Discussion

The dual representation transforms the superhedging problem from a dynamic trading problem into a static system of linear inequalities. The first formulation uses consistent price systems and is closely related to martingale pricing theory. The second formulation, based on vector probability measures, connects the problem to convex analysis and set-valued risk measures.

This duality plays a crucial role in both theoretical analysis and numerical computation, as it allows the use of optimization techniques to characterize and compute the set of superhedging portfolios.

9 Recursive Construction

The set of superhedging portfolios can be computed recursively by working backward through the event tree.

9.1 Terminal Condition

At the final time T , we have:

$$SHP_T(X)(\omega) = X(\omega) + K_T(\omega).$$

Interpretation: At maturity, we only need to cover the payoff $X(\omega)$. The addition of the solvency cone $K_T(\omega)$ accounts for all portfolios that can be adjusted into $X(\omega)$ through feasible trades.

9.2 Backward Recursion

For earlier times, the sets are computed as:

$$SHP_t(X)(\omega) = \bigcap_{\bar{\omega} \in \text{succ}(\omega)} (SHP_{t+1}(X)(\bar{\omega}) + K_t(\omega)).$$

Interpretation: At each time step, we choose portfolios that remain feasible for all possible future states. The intersection ensures that the strategy works under every possible evolution of the market, while the addition of $K_t(\omega)$ allows for adjustment through trading at the current time.

9.3 Our Understanding

From this recursive structure, we observe that superhedging is inherently a dynamic and robust problem. At each step, the portfolio must be chosen in a way that is safe against all future uncertainties.

This backward construction highlights an important idea:

Superhedging is achieved by combining feasible trades today with robustness against all possible future scenarios.

10 Conclusion

In this report, we studied how transaction costs fundamentally change the structure of financial markets. Instead of working with prices alone, we modeled the market through solvency cones, which capture all feasible trades.

We observed that superhedging is no longer a scalar problem, but a set-valued one. This leads to a richer and more realistic framework for hedging under market frictions.

The recursive construction highlights how hedging decisions must account for all possible future scenarios, making the approach inherently robust.

Overall, this framework provides both a geometric and probabilistic understanding of markets with transaction costs.